Benchmark Failure Under Deployment Frictions: When Forecast Rankings Misselect Operational Winners

Abstract

Forecasting systems are often selected using forecast-side scores, but under deployment frictions these scores can mis-rank systems relative to realized operational performance. We identify a benchmark failure mode: under a fixed deployed interface, forecast-side ranking need not preserve deployed ranking once proposed actions are not directly realized, creating recurrent model-selection error. We separate two evaluation questions—same proposal under different interfaces (Q1) and different forecasters under a fixed interface (Q2)—and study them in synthetic, inventory, and forecasting-oriented support settings. Across controlled and operational evidence, zero-friction alignment is exact or near-stable, while moderate-to-high friction induces repeated ranking reversals and deployed misselection. These results support a narrow but consequential claim: forecasting systems under frictions should be benchmarked with realized-action-aware evaluation criteria rather than forecast-side quantities alone.

1 Introduction

A forecasting benchmark is unreliable for deployment if the model that wins on forecast-side scores is recurrently not the model that wins after actions are realized under friction. Forecasting systems are often evaluated with forecast-side metrics or target-side decision summaries, which can be reasonable when proposed actions are treated as if they were directly realized. Under deployment frictions, however, forecasts pass through interfaces that alter realized behavior, costs, and downstream quality, so forecast-side ranking can cease to be order-preserving for deployed performance.

This paper studies two evaluation questions for forecasting-driven systems under frictions. Q1 isolates the mechanism by which realized outcomes depart from target-side or forecast-side quantities: if the same frozen proposal path is held fixed, can different interfaces produce different realized outcomes? Q2 asks the benchmark-relevant selection question: under one deployed rule, can forecast-side ranking still be trusted to select the operationally best system? These are evaluative rather than predictive questions; the goal is not to propose a stronger forecaster, but to make benchmark interpretation and model selection more reliable once forecasts pass through constrained action layers.

This paper makes three narrow contributions. First, it identifies a benchmark failure mode for forecasting systems under frictions: under a fixed deployed interface, forecast-side ranking can fail to preserve deployed ranking. Second, it separates mechanism (Q1) from selection consequence (Q2), clarifying two evaluation questions that are often conflated in forecasting-driven systems. Third, it shows across controlled and operational evidence that ranking drift is not merely descriptive: at moderate-to-high friction, it produces recurrent deployed misselection. We support this claim with controlled synthetic evidence, one operational inventory domain, and exact-control corroboration from portfolio Q1 support.

2 Two Evaluation Questions

Let $\hat{y}_t$ denote a forecast, a_t^{tar} a proposed target action derived from that forecast, and a_t^{exec} the realized action after an interface applies under friction. Let $U(a, y)$ denote a realized operational score evaluated against the realized outcome y_t. We distinguish three evaluation objects: forecast-side quality, target-side decision quality, and realized-action quality.

Q1 (same proposal, different interface). Holding the proposal fixed at $a_{1:T}^{tar}$, we ask whether varying only the interface that maps proposals to realized actions changes realized operational outcomes. This is an exact-control question: proposal identity must be locked, so any downstream difference must arise from the interface rather than from a different forecast or proposal source.

Q2 (fixed interface, different forecasts). Holding the interface fixed, we ask whether varying the forecaster changes the relationship between forecast-metric ranking and deployed operational ranking. This is not an evaluation-object mismatch test in the Q1 sense. Instead, it asks whether forecast-quality ranking and deployed-quality ranking remain aligned once forecasting outputs are judged through one fixed operational rule.

These questions are complementary but not coequal in evidentiary role. Q1 isolates why realized-action evaluation can depart from target-side or forecast-side quantities under a fixed proposal source, while Q2 centers the practical benchmark claim about ranking drift and model selection under a fixed interface. We do not combine them into one omnibus claim because they answer different evaluation questions and play different roles in the evidence hierarchy.

3 Evidence Design

Our evidence design separates a zero-friction sanity anchor, operational consequence, and support-only case evidence. Synthetic serves as the zero-friction sanity anchor: it establishes clean alignment at zero friction and controlled ranking drift under positive friction. Inventory provides operational near-alignment evidence rather than the exact sanity anchor, and it is where ranking drift becomes a deployed model-selection consequence under holding, stockout, and change-cost tradeoffs. Portfolio provides exact-control corroboration for Q1. The paper's practical benchmark claim is centered on Q2: whether forecast-side ranking can be trusted

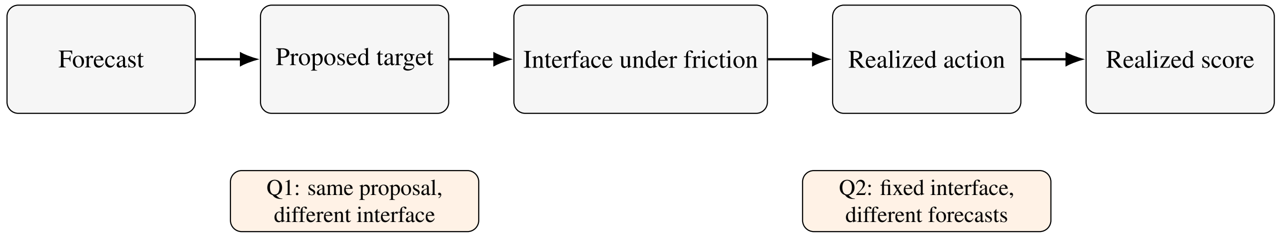

Figure 1: Forecasts are converted into proposed target actions and then translated into realized actions through an interface operating under friction, which creates two distinct evaluation questions: whether the same frozen proposal can yield different realized outcomes under different interfaces (Q1), and whether forecast-metric ranking matches deployed operational ranking under a fixed interface (Q2). This separation changes the evaluation object and can therefore alter benchmark interpretation and model-selection stability under frictions.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.70	0.022	3/10
0.25	Small MLP	Linear AR	0.70	0.053	3/10
0.50	Small MLP	Moving average (7)	0.10	0.308	9/10
1.00	Small MLP	Moving average (7)	0.00	1.187	10/10

Table 1: Forecast-side vs. deployed-side model selection under a fixed replenishment interface. Winner columns report the most frequent seed-level best model at each friction level. Positive mean deployed gap means the forecast-selected model underperforms the deployed-selected model by that amount. Deployed-suboptimal seeds / total reports the number of seeds in which the forecast-selected model is not a deployed best model. Table 1 shows the operational selection consequence in the required inventory domain; Figure 2 retains the cross-domain ranking-drift pattern in both synthetic and inventory.

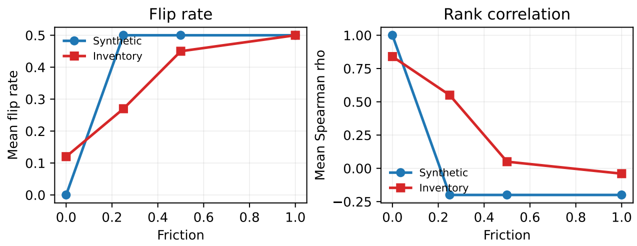

Figure 2: Q2 — under a fixed interface, forecast-side ranking can misselect operational winners under friction. Synthetic provides the zero-friction sanity anchor, while inventory shows the operational selection consequence. Ranking drift is interesting, but recurrent deployed misselection is the benchmark consequence: Figure 2 shows the cross-domain ranking pattern, while Table 1 shows the inventory-side selection consequence.

for deployed model selection under a fixed operational rule. Q1 clarifies the mechanism by which realized-action evaluation departs from target-side or forecast-side quantities. Synthetic and inventory remain the required headline evidence, while the appendix adds fit-oriented event-forecasting and load-following support.

In inventory Q2, the locked main table compares five forecasting families under one responsive replenishment rule: Naive persistence, Moving average (7), Linear AR, Small MLP, and Small GRU. To reduce baseline-specification ambiguity, the appendix additionally reports a broader stronger-baseline block with validation-selected family representatives under the same split, held-out protocol, validation criterion, and standardized two-stage tuning procedure; simple heuristic baselines remain fixed ex ante. The main text retains the locked five-family evidence, while the appendix records the broader family sweep, candidate grids, and selection outcomes for transparency rather than as a replacement leaderboard.

4 Results

4.1 Q2: Fixed interface, benchmark failure and selection consequence

Q2 is the paper's headline question: under one fixed deployed interface, can forecast-side ranking still be trusted for deployed model selection? Synthetic serves as the zero-friction sanity anchor. In the controlled benchmark, zero-friction rankings align exactly, while positive friction produces repeated pairwise flips and lower rank correlation, with mean flip rate rising from 0.000 to 0.500.

Inventory carries the operational consequence. The friction 0.25 row is best read as a transition regime: the most frequent deployed winner changes to Linear AR, but agreement remains 0.70, the deployed-suboptimal count remains 3/10, and the median deployed gap remains 0.0. By friction 0.5, however, benchmark failure is already recurrent: the forecast-selected model is deployed-suboptimal in 9/10 seeds, agreement falls to 0.10, and the median deployed gap reaches 0.227. At friction 1.0, deployed suboptimality rises to 10/10 seeds, with mean and median deployed gaps of 1.187 and 1.162, respectively. Ranking drift is interesting, but recurrent deployed misselection is the benchmark consequence.

Appendix Tables A3–A6 widen the inventory comparison set with stronger classical and neural families under a standardized two-stage tuning protocol for baseline-specification transparency. A more reactive appendix follow-up strengthens the moderate-to-high-friction mismatch pattern, but zero-friction agreement remains only partial, so we still treat the broader-family sweep as appendix defense rather than as additional main-text confirmation.

To reconnect this operational result to more canonical forecasting language, we also evaluate a minimal event-forecasting setting in which models emit event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. In that appendix support domain, agreement is 0.75 at friction 0.00, falls to 0.30 by 0.50, and reaches 0.05 at 1.00. Over the same range, the deployed winner shifts from Reactive sharp to Lagged smoother.

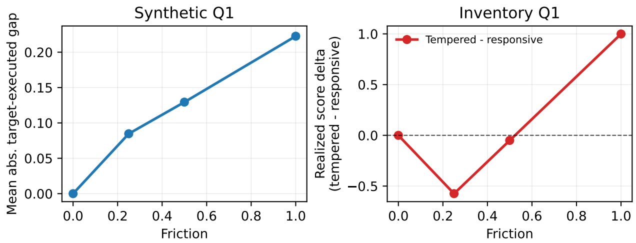

Figure 3: Q1 — the same proposal can yield different realized outcomes under different interfaces. At zero friction, target-side and realized-side quantities coincide or nearly coincide. Under positive friction, interface-mediated differences emerge. The synthetic benchmark shows the controlled phenomenon cleanly, while the inventory domain shows a threshold pattern: low friction can remain mixed, but sufficiently high friction makes the tempered interface beneficial through reduced adjustment costs. The two panels use different y-axis quantities because the synthetic and inventory Q1 evidence emphasize controlled gap emergence and operational score effects, respectively.

4.2 Q1: Same proposal, different interface

Q1 explains why the benchmark failure in Q2 can arise. In the synthetic benchmark, holding the proposal fixed while varying only the interface yields exact zero-friction coincidence and a proposal-versus-realization gap that rises from 0.000 at zero friction to 0.223 at friction 1.0. This establishes the mechanism cleanly: once frictions intervene, realized-action evaluation can depart from target-side or forecast-side quantities even when the proposal path is unchanged.

Inventory reproduces the same mechanism in an operational setting. The zero-friction row is exact by construction, while positive friction shows a threshold story rather than a monotone one: low friction can still favor responsiveness, but sufficiently high friction makes tempered execution beneficial, with the tempered-minus-responsive delta reaching 0.999 at friction 1.0. Portfolio then provides exact-control corroboration: proposal identity is verified directly, zero-cost behavior is near-flat in $7/9$ seed-universe groups, and positive-cost improvements recur in $3/3$ universes. Q1 therefore explains why realized-action evaluation can depart from target-side or forecast-side objects, while Q2 shows the benchmark consequence when model selection relies on forecast-side ranking.

5 Discussion and Limitations

The paper's practical benchmark claim is centered on Q2: under a fixed deployed interface, forecast-side ranking can fail to preserve deployed ranking once frictions separate proposed actions from realized outcomes. This is a benchmark failure mode, not a claim that forecast metrics are useless or that one interface is uniformly superior. Q1 clarifies the mechanism: once realized actions depart from proposed targets, forecast-side, target-side, and realized-action quality are no longer the same object.

The claim is intentionally narrow. Synthetic provides the zero-friction sanity anchor, inventory provides the main operational evidence, and portfolio contributes exact-control Q1 support rather than balanced Q1/Q2 support. Event micro and load-following remain appendix support only, and the stronger-baseline appendix block remains transparency and baseline-specification defense rather than additional headline confirmation: a more reactive follow-up strengthens the moderate/high-friction mismatch pattern, but its zero-friction agreement remains only partial relative to the paper's sanity-anchor standard. The claim is narrow in scope but direct in consequence: if forecast-side evaluation recurrently selects the wrong deployed winner under friction, then benchmark conclusions and model choice can both be wrong even when forecast quality itself is measured correctly.

The resulting implication is evaluative rather than predictive: forecasting systems under deployment constraints should be judged with realized-action-aware criteria rather than forecast-side quantities alone.

A Evidence Hierarchy

Domain	Question	Fixed	Varying	Zero-fric.	Positive-fric.	Role
Synthetic	Q1	proposal	interface	exact agreement	mismatch emerges	required
Synthetic	Q2	interface	forecaster	rank alignment	ranking flips	required
Inventory	Q1	proposal	interface	coincide at zero	threshold story	required
Inventory	Q2	interface	forecaster	low mismatch	persistent drift	required
Portfolio	Q1	proposal	interface	mostly near-flat	positive-cost recurrence	support
Event micro	Q2	interface	forecaster	aligned at zero / very low	mixed then ranking drift	appendix support

Table A1: Evidence map for the two evaluation questions. Q1 and Q2 are separated by construction. Synthetic establishes the controlled phenomenon, inventory provides one operational domain, portfolio contributes support-only exact-control evidence for Q1, and the appendix adds a fit-oriented minimal event-forecasting Q2 support row.

B Additional Q2 Package Details

The additional Q2 package is descriptive rather than headline evidence. It summarizes the same-interface rank-based package after the fifth inventory baseline is added, keeps the synthetic required-domain result visible, and records why portfolio is not part of the headline Q2 claim.

Portfolio failed the zero-friction sanity gate and is therefore excluded from headline Q2 evidence. Inventory remains near-aligned at zero friction but no longer passes the older appendix rank gate once the fifth baseline is added, which is why the main text now centers the inventory selection-drift table rather than this appendix summary.

C Stronger Inventory Q2 Baselines

To reduce baseline-specification ambiguity, we widen the inventory Q2 comparison set with stronger classical and neural families while leaving the environment, split, friction grid,

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	pass	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed gate
Inventory	required	fail	10	0.120	0.840	1.0000	0.500	Small MLP / Moving average (7)	1.000	flip > 0.10
Portfolio	stretch	excluded	3	0.611	-0.200	0.0005	0.778	EWMA (20) / Rolling mean (20)	1.000	flip > 0.10; rho < 0.50

Table A2: Additional paper-facing Q2 package summary. The appendix records the same-interface rank-based package after the fifth inventory baseline is added, keeps the synthetic required-domain result visible, and keeps portfolio out of the headline Q2 claim.

fixed responsive interface, deployed metric, and tie policy unchanged. All tunable families share the same held-out protocol, validation criterion, and standardized two-stage tuning structure, while the two simple heuristics remain fixed ex ante. Family representatives are selected by validation negative MAE only, and the held-out forecast-side winners in the robustness table are then computed from the same held-out forecast metric used in the locked inventory Q2 pipeline.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Gradient-boosted trees	Regularized linear + lag search	0.50	0.027	5/10
0.25	Gradient-boosted trees	Regularized linear + lag search	0.80	0.042	2/10
0.50	Gradient-boosted trees	Gradient-boosted trees	0.80	0.072	2/10
1.00	Gradient-boosted trees	Gradient-boosted trees	0.60	0.170	4/10

Table A5: Expanded-family inventory Q2 robustness summary. This appendix block is reported for transparency and baseline-specification defense rather than as additional headline evidence.

In this broader appendix block, zero-friction agreement is 0.50, the 0.25 and 0.5 rows remain comparatively aligned, and the strongest mean deployed gap is 0.170 at friction 1.0. We therefore keep this first block as transparency and defense evidence against baseline-specification artifacts rather than treat it as an additional headline confirmation of recurrent misselection.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A6: Reactive follow-up for the expanded-family inventory Q2 block. This follow-up widens the search toward shorter-lag and higher-capacity representatives while preserving the same split, interface, friction grid, and held-out protocol.

A more reactive appendix follow-up strengthens the moderate-to-high-friction pattern without changing the locked evaluation protocol. Agreement is 0.60 at friction 0.00, remains mixed at 0.25, falls to 0.50 at 0.50, and reaches 0.10 at 1.00; over the same rows, deployed-suboptimal counts rise from $4/10$ to $9/10$, and the deployed winner shifts to Moving average (7) at the highest friction level. Because the zero-friction row remains only partially aligned and the 0.25 row is still mixed, we treat this follow-up as appendix robustness rather than headline confirmation.

D Minimal Event-Forecasting Micro-Benchmark

We additionally construct a minimal binary-event setting in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. This appendix artifact is included as appendix-level support rather than headline evidence: its role is a fit-oriented minimal confirmation of ranking drift under switching friction, not a new benchmark centerpiece. Brier remains the canonical forecast-side criterion, while log loss is recorded only as a robustness diagnostic and does not determine winners.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap
0.00	Reactive sharp	Reactive sharp	0.75	0.002
0.05	Reactive sharp	Reactive sharp	0.70	0.002
0.10	Reactive sharp	Calibrated baseline	0.55	0.002
0.25	Reactive sharp	Calibrated baseline	0.45	0.004
0.50	Reactive sharp	Calibrated baseline	0.30	0.013
1.00	Reactive sharp	Lagged smoother	0.05	0.061

Table A7: Forecast-side vs. deployed-side model selection in a minimal event-forecasting micro-benchmark. Under a fixed thresholding interface with switching friction, the forecast-selected model and the deployed-selected model remain broadly aligned at zero and very low friction before diverging recurrently as friction increases.

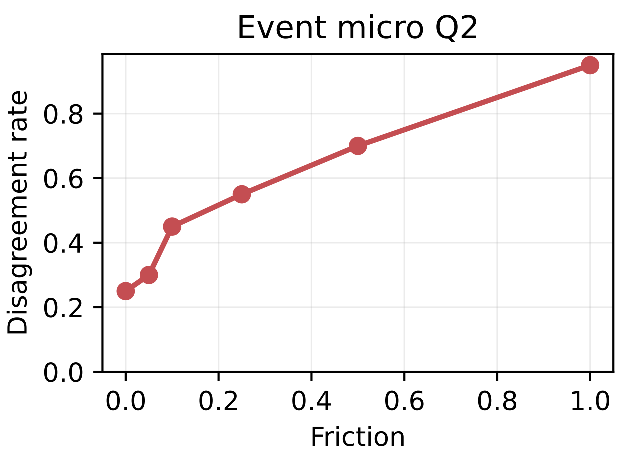

Figure A1: Ranking mismatch in a minimal event-forecasting micro-benchmark. Disagreement between forecast-side and deployed-side model selection is lowest at zero and very low friction and increases as switching frictions rise.

At zero and very low switching friction, forecast-side and deployed-side selection remain broadly aligned in this domain. As switching friction rises, disagreement increases, intermediate rows become mixed, and the deployed-side winner shifts from the reactive forecast-side winner toward more stable models, culminating in a lagged-smoother winner at the highest friction row. This is the same qualitative ranking-

Family	Input	Stage 1	Stage 2	Seeds	Representative rule
Naive persistence	fixed heuristic	–	–	final 10	fixed ex ante
Moving average (7)	fixed heuristic	–	–	final 10	fixed ex ante
Linear AR	legacy tabular	5	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small MLP	legacy tabular	8	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Small GRU	sequence (lookback 28)	4	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Regularized linear + lag search	lagged tabular	18	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Gradient-boosted trees	lagged tabular	64	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
Large MLP	legacy tabular	32	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config
GRU variant	sequence	24	top 3	s1 0,1 / s2 0,1,2 / final 0-9	best mean val. metric, then median, then pre-registered simpler config

Table A3: Standardized tuning protocol for the stronger-baseline transparency block. All tunable families share the same split, validation criterion, and two-stage selection structure; the two simple heuristics remain fixed ex ante.

Family	Representative	Validation mean	Validation median
Naive persistence	fixed ex ante	-2.911	-2.981
Moving average (7)	fixed ex ante	-3.265	-3.251
Linear AR	alpha=0.0001	-2.449	-2.512
Small MLP	lr=0.001, wd=0.0, batch=32	-2.331	-2.390
Small GRU	lr=0.001, batch=32	-2.628	-2.571
Regularized linear + lag search	lag=28, elasticnet, alpha=0.1	-2.501	-2.611
Gradient-boosted trees	lag=28, trees=100, depth=2, lr=0.05, leaf=10	-2.368	-2.362
Large MLP	w=128, d=3, drop=0.0, lr=0.001, wd=1e-05	-2.053	-2.078
GRU variant	L=28, h=64, layers=1, drop=0.0, lr=0.001	-2.628	-2.736

Table A4: Validation-selected representatives for the broader inventory Q2 family sweep. Family representatives are chosen by validation negative MAE only; deployed metrics do not enter family selection.

drift pattern emphasized in the main Q2 discussion, but in a deliberately smaller event-forecasting domain.

E Load-Following Support Domain

Using UCI ElectricityLoadDiagrams20112014, we build an aggregate multi-client load-following domain by summing disjoint client groups and evaluating at the selected 60-minute operational resolution. Groups 0 and 1 are used only for restricted calibration, while groups 2–9 serve as held-out evaluation units, so this appendix replaces the earlier household proxy with a stronger cross-sectional operational design.

A grouping-only balance-repair rerun substantially reduces group imbalance and Q1 target clipping while preserving the Q2 drift structure. A final restricted cap/margin search, however, yields no jointly feasible configuration that both preserves the Q2 promotion gate and satisfies the Q1 requirement. The domain therefore remains appendix-only support.

Domain	Role	Groups	Res.	Q1	Q2	Balance	Note
Aggregate electricity-load proxy	support	8	60min	mixed support	promotion pass	balance_warn	appendix support only

Table A8: Load-following support-domain summary. This appendix-only domain remains outside the paper's required evidence hierarchy in the current submission round.

A grouping-only balance repair reduces imbalance and lowers the held-out Q1 target-clip rate below the gate threshold while preserving the Q2 drift structure. A final restricted cap/margin search then finds no jointly feasible configuration that preserves the Q2 gate while satisfying the remaining Q1 requirement.

Under a fixed responsive interface, Q2 provides strong operational support in this domain, but the domain remains appendix-only in the current submission round. A grouping-only balance repair substantially reduces imbalance and lowers the held-out Q1 target-clip rate, showing that clipping was partly imbalance-driven, yet Q1 high-friction win-rate remains mixed. A final restricted cap/margin search then finds no jointly feasible configuration that preserves the Q2 gate while satisfying the remaining Q1 condition. This appendix evidence should therefore be read as additional operational support, not as part of the paper's required evidence hierarchy.